Graphs and Transformations

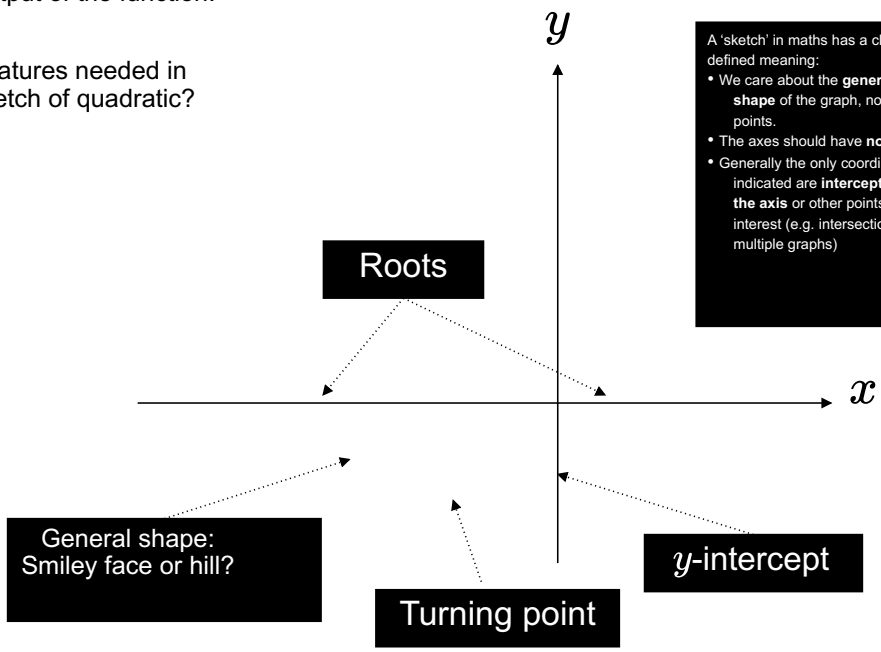
Chapter 2 · Original slide 26

4: Quadratic Graphs

26

Recall that x refers to the input of a function, and the expression $f(x)$ refers to the output. For graph sketches, we often write $y = f(x)$, i.e. we set the y values to be the output of the function.

Features needed in sketch of quadratic?



A 'sketch' in maths has a clearly defined meaning:
- We care about the **general shape** of the graph, not exact points.
- The axes should have **no scale**.
- Generally the only coordinates indicated are **intercepts with the axis** or other points of interest (e.g. intersections of multiple graphs)

Chapter 2 · Original slide 27

Example

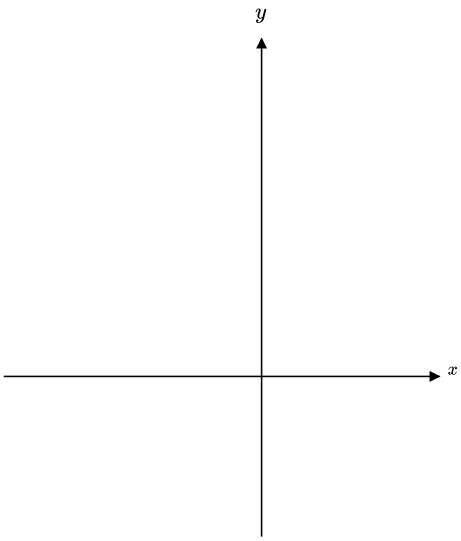
27

Sketch the graph of $y = x^2 + 3x - 4$ and find the coordinates of the turning point.

Roots:

y-intercept:

Turning point:



Recall that if $f(x) = (x + a)^2 + b$, the minimum output is b and $-a$ is the x value which minimises it. i.e. Turning point is $(-a, b)$

Graphs and Transformations

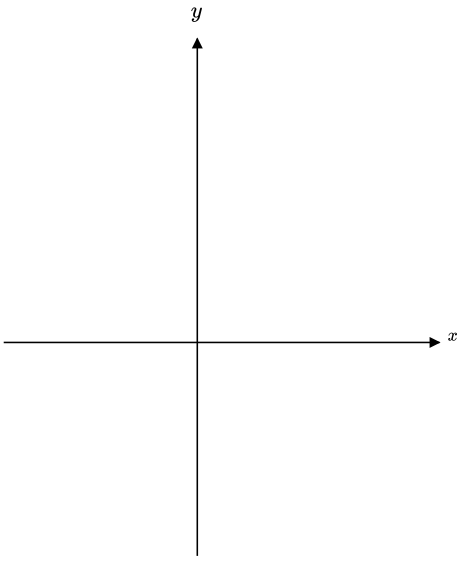
Chapter 2 · Original slide 28

Example

28

Sketch the graph of $y = 4x - 2x^2 - 3$ and find the coordinates of the turning point. Write down the equation of the line of symmetry.

Roots:



y-intercept:

Turning point:

Line of reflection:

Chapter 2 · Original slide 29

Test Your Understanding

29

Sketch the following, indicating any intercepts with the axis, the turning point and the equation of the line of symmetry.

a $y = x^2 + 4$

c $y = 5x + 3 - 2x^2$

b $y = x^2 - 7x + 10$

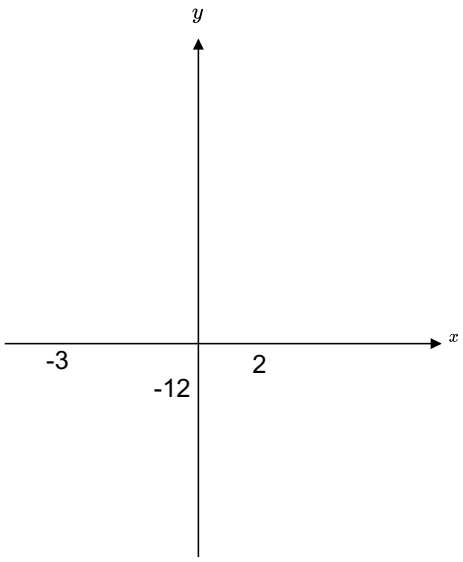
d $y = x^2 + 4x + 11$

Graphs and Transformations

Chapter 2 · Original slide 30

Determining the Equation using a Graph

30

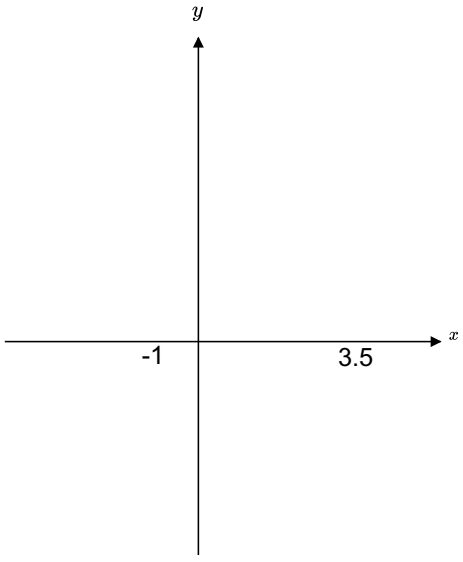


Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$.

Chapter 2 · Original slide 31

Further Example

31



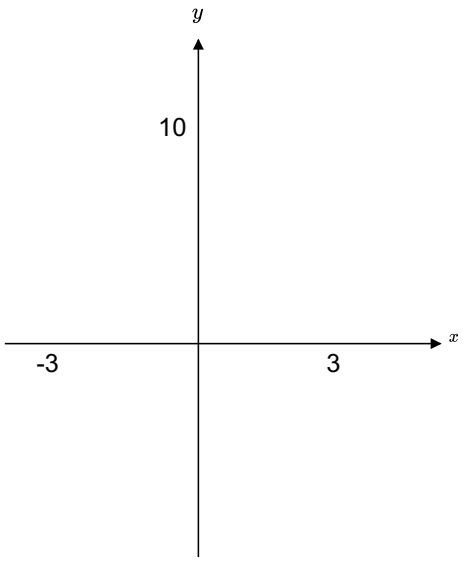
Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$, where a, b, c are integers.

Graphs and Transformations

Chapter 2 · Original slide 32

Test Your Understanding

32



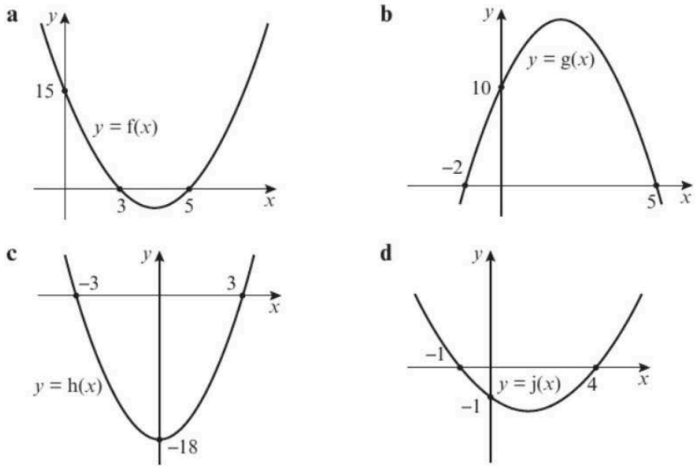
Determine an equation of this quadratic graph.

Chapter 2 · Original slide 33

Exercise 2F Pure Y1 Page 30

33

2 These sketches are graphs of quadratic functions of the form $ax^2 + bx + c$. Find the values of a, b, c for each function.



Graphs and Transformations

Chapter 2 · Original slide 34

Exercise 2F Pure Y1 Page 30

34

3 The graph of $y = ax^2 + bx + c$ has a minimum at $(5, -3)$ and passes through $(4, 0)$.
Find the values of a, b, c.

Extension Question:
[MAT 2003 1H] Into how many regions is the plane divided when the following three parabolas are drawn?
$y = x^2$
$y = x^2 - 2x$
$y = x^2 + 2x + 2$

y

x

Chapter 4 · Original slide 4

Polynomial Graphs

4

In Chapter 2 we briefly saw that a **polynomial** expression is of the form:

$$a + bx + cx^2 + dx^3 + ex^3 + \dots$$

where $a, b, c, d, e, \dots$ are constants (which could be 0).

The **order** of a polynomial is its highest power.

Order	Name
0	
1	
2	
3	
4	
5	

These are covered in Chapter 5.

Chapter 2 explored the graphs for these.

We will cover these now.

While these are technically beyond the A Level syllabus, we will look at how to sketch polynomials in general.

Graphs and Transformations

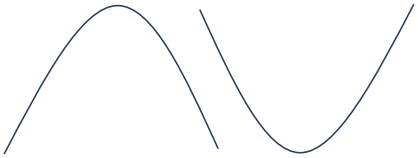
Chapter 4 · Original slide 5

Polynomial Graphs

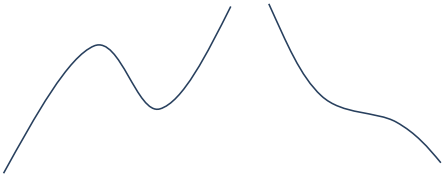
5

Order:

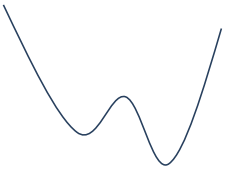
2



3



4



What property connects the order of the polynomial and the shape?

Bro Note: …Actually this is not strictly true, e.g. consider $y = x^4$, which has a U shape. But this is because multiple turns are being squashed into a single point.

In Chapter 2 how did we tell what way up a quadratic is, and why does this work?

Chapter 4 · Original slide 6

Polynomial Graphs

6

e.g. If $y = 2x^2 + 3$, try a large positive value like $x = 1000$. We can see we'd get a large positive y value. Thus as $x \to \infty$, $y \to \infty$

Equation	If $a > 0$	Resulting Shape	If $a < 0$	Resulting Shape
$y = ax^2 + bx + c$	As $x \to \infty, y \to \infty$ As $x \to -\infty, y \to \infty$		As $x \to \infty, y \to -\infty$ As $x \to -\infty, y \to -\infty$	
$y = ax^3 + bx^2 + cx + d$				
$y = ax^4 + bx^3 + cx^2 + dx + e$				
$y = ax^5 + bx^4 + \ldots$				

If a>0, what therefore can we say about the shape if:
• The order is odd:
• The order is even:

Graphs and Transformations

Chapter 4 · Original slide 7

Cubics

7

Sketch the curve with equation
$y = (x-2)(1-x)(1+x)$

Features you must consider:

Shape?

Roots?

y-intercept?

y

x

Tip: It's incredibly easy to forget to write in one of the intercepts. So don't!

Sketch the curve with equation
$y = x^2(x-1)$

Shape?

Roots?

y-intercept?

y

x

Chapter 4 · Original slide 8

Cubics

8

Sketch the curve with equation
$y = (2-x)(x+1)^2$

Shape?

Roots?

y-intercept?

y

x

Sketch the curve with equation
$y = (x-4)^3$

Shape?

Roots?

y-intercept?

y

x

Exam Notes: The term 'point of inflection' has been removed from the new A Level syllabus (bad idea!!).

A point of inflection is where the curve goes from 'convex' to 'concave' (or vice versa), i.e. curves in one direction before and curves in another direction after. You might have encountered these terms in Physics.

Graphs and Transformations

Chapter 4 · Original slide 9

Cubics with Limited Roots

9

Sketch the curve with equation

$$y = (x + 1)\left(x^2 + x + 1\right)$$

Shape?

Roots?

y-intercept?





However, in Chapter 12, we'll be able to work **turning points** using 'differentiation', and hence conclude that it doesn't have any!

Chapter 4 · Original slide 10

Finding the equation yourself

10

Edexcel C1 May 2013(R) Q9

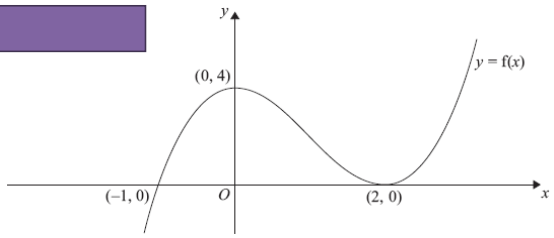


Figure 1 shows a sketch of the curve C with equation $y = f(x)$.
The curve C passes through the point (–1, 0) and touches the x-axis at the point (2, 0).
The curve C has a maximum at the point (0, 4).
The equation of the curve C can be written in the form.

$$y = x^3 + ax^2 + bx + c$$

where a, b and c are integers.
(a) Calculate the values of a, b, c.

Graphs and Transformations

Chapter 4 · Original slide 11

Test Your Understanding

11

1 Sketch the curve with equation $y = x(x-3)^2$

y

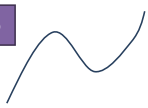
x

2 Sketch the curve with equation $y = -(x+2)^3$

y

x

3



A curve has this shape, touches the x axis at 3 and crosses the x axis at -2. Give a suitable equation for this graph.

Sketch the curve with equation $y = 2x^2(x-1)(x+1)^3$

Chapter 4 · Original slide 12

Exercise 4A –Y1 Pure pages 63

12

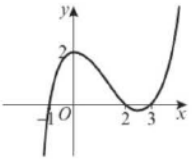
5 The graph of $y = x^3 + bx^2 + cx + d$ is shown opposite, where b, c and d are real constants.

a Find the values of b, c and d.

b Write down the coordinates of the point where the curve crosses the y-axis.



6 The graph of $y = ax^3 + bx^2 + cx + d$ is shown opposite, where a, b, c and d are real constants.
Find the values of a, b, c and d.



Graphs and Transformations

Chapter 4 · Original slide 13

Exercise 4A –Y1 Pure pages 63

13

7 Given that $f(x) = (x-10)(x^2-2x)+12x$

a Express $f(x)$ in the form $x(ax^2+bx+c)$ where a, b, c are real constants.

b Hence factorise $f(x)$ completely.

c Sketch the graph of $y = f(x)$ showing clearly the points where the graph intersects the axes.

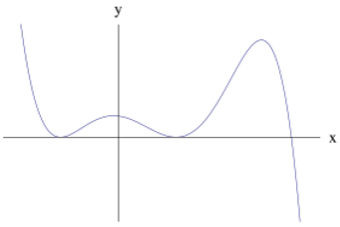
Chapter 4 · Original slide 14

Exercise 4A Extension

14

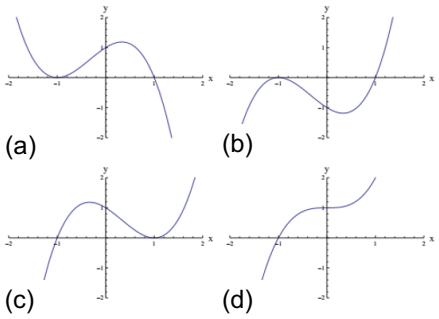
Extension

1 [MAT 2012 1E] Which one of the following equations could possibly have the graph given below?



A) $y = (3-x)^2(3+x)^2(1-x)$
B) $y = -x^2(x-9)(x^2-3)$
C) $y = (x-6)(x-2)^2(x+2)^2$
D) $y = (x^2-1)^2(3-x)$

2 [MAT 2011 1A] A sketch of the graph $y = x^3 - x^2 - x + 1$ appears on which of the following axis?



Cubics can sometimes be factorised by pairing the terms:

Graphs and Transformations

Chapter 4 · Original slide 15

Recap

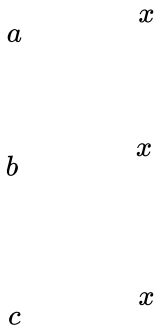
15

If we sketched $y = (x-a)(x-b)^2(x-c)^3$ what happens on the x-axis at:

$x = a$:

$x = b$:

$x = c$:



Chapter 4 · Original slide 16

Quartics

16

If you understand the principle of sketching polynomials in general, then sketching quartics shouldn't feel like anything new.
Recall that if the x^4 term is positive, the 'tails' both go upwards, otherwise downwards.

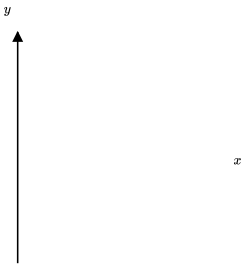
Sketch the curve with equation
$y = x(x+1)(x-2)(x-3)$

Shape:
Roots:
y-intercept:



Sketch the curve with equation
$y = (x-2)^2(x+1)(3-x)$

Shape:
Roots:
y-intercept:



Graphs and Transformations

Chapter 4 · Original slide 17

Quartics

17

Sketch the curve with equation
$y = (x+1)(x-1)^3$

-1 root only appears once so line crosses at $x = -1$
+1 root triple repeated so point of inflection at $x = 1$



Sketch the curve with equation
$y = (x-2)^4$

2 is a quadruple repeated root! Because the line effectively crosses the axis 4 times all at -2, it ends up in the opposite direction, and hence looks like a 'touch' point.

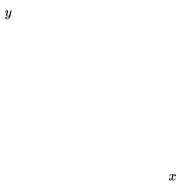


Chapter 4 · Original slide 18

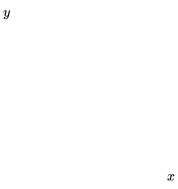
Test Your Understanding

18

Sketch the curve with equation
$y = x^2(x+1)(x-1)$



Sketch the curve with equation
$y = -(x+1)(x-3)^3$



Graphs and Transformations

Chapter 4 · Original slide 19

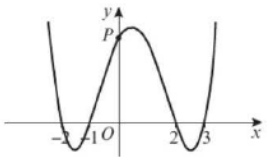
Exercise 4B – Pure Y1 Pages 65-66

19

3 The graph of $y = x^4 + bx^3 + cx^2 + dx + e$ is shown opposite, where b, c, d and e are real constants.

a Find the coordinates of point P. **(2 marks)**

b Find the values of b, c, d and e. **(3 marks)**



4 Sketch the graph of $y = (x + 5)(x - 4)(x^2 + 5x + 14)$.

Chapter 4 · Original slide 20

Exercise 4B Extension

20

Extension

1 *[STEP I 2012 Q2a]*
a. Sketch $y = x^4 - 6x^2 + 9$
b. For what values of b does the equation $y = x^4 - 6x^2 + b$ have the following number of distinct roots (i) 0, (ii) 1, (iii) 2, (iv) 3, (v) 4.

a)

b) By changing b, we shift the graph up and down. Then we can see that:

i) 0 roots:
ii) 1 root:
iii) 2 roots:
iv) 3 roots:
v) 4 roots:

Graphs and Transformations

Chapter 4 · Original slide 21

GCSE RECAP: Reciprocal Graphs

21

Sketch $y = \frac{1}{x}$

y

x

Sketch $y = -\frac{3}{x}$

y

x

Note: The scaling caused by the 3 isn't observable for this graph in isolation because the axes have no scale. This will only be observation for multiple graphs on the same axes.

Chapter 4 · Original slide 22

Reciprocal Graphs

22

Sketch $y = \frac{3}{x^2}$

This is new to the A Level 2017 syllabus.

y

x

Sketch $y = -\frac{4}{x^2}$

y

x

Hint: Note that anything squared will always be at least 0.

Graphs and Transformations

Chapter 4 · Original slide 23

Reciprocal Graphs

23

On the same axes, sketch $y = \frac{1}{x}$ and $y = \frac{3}{x}$

y $y = \frac{3}{x}$

The y value for $y = \frac{3}{x}$ will be 3 times greater than $y = \frac{1}{x}$

$y = \frac{1}{x}$ x

Chapter 4 · Original slide 24

Exercise 4C – Pure Y1 page 67

24

1 Use a separate diagram to sketch each pair of graphs.

a $y = \frac{2}{x}$ and $y = \frac{4}{x}$ **b** $y = \frac{2}{x}$ and $y = -\frac{2}{x}$ **c** $y = -\frac{4}{x}$ and $y = -\frac{2}{x}$

d $y = \frac{3}{x}$ and $y = \frac{8}{x}$ **e** $y = -\frac{3}{x}$ and $y = -\frac{8}{x}$

Graphs and Transformations

Chapter 4 · Original slide 25

Exercise 4C – Pure Y1 page 67

25

2 Use a separate diagram to sketch each pair of graphs. **a** $y = \frac{2}{x^2}$ and $y = \frac{5}{x^2}$ **b** $y = \frac{3}{x^2}$ and $y = -\frac{3}{x^2}$ **c** $y = -\frac{2}{x^2}$ and $y = -\frac{6}{x^2}$

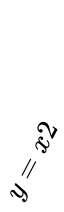
Chapter 4 · Original slide 34

Transformations of Functions

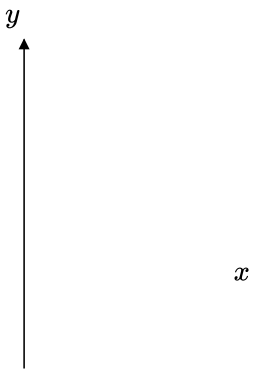
34

Suppose $f(x) = x^2$

Sketch $y = f(x)$:



Sketch $y = f(x + 2)$



What do you notice about the relationship between the graphs of $y = f(x)$ and $y = f(x + 2)$?

Graphs and Transformations

Chapter 4 · Original slide 35

Transformations of Functions

35

This is all you need to remember when considering how transforming your function transforms your graph...

	Affects which axis?	What we expect or opposite?
Change **inside** $f()$		
Change **outside** $f()$		

Therefore...

$y = f(x - 3)$ ➡

$y = f(x) + 4$ ➡

$y = f(5x)$ ➡

$y = 2f(x)$ ➡

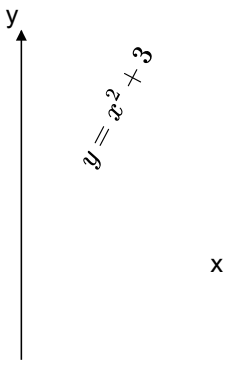
Chapter 4 · Original slide 36

Sketching transformed graphs

36

Sketch $y = x^2 + 3$

Imagine a sketch of $y = x^2$ and then do the translation, ensuring you adjust any intercepts with the axes.



Sketch $y = \frac{2}{x+1}$

y

The transformation might result in new intercepts or roots. You can find these in the usual way. **Do not forget them!**

x

Draw asymptotes using a dotted line and write its equation on it.

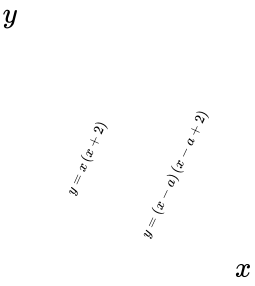
Graphs and Transformations

Chapter 4 · Original slide 37

More Examples

37

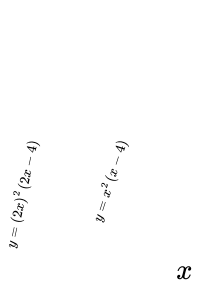
Sketch $y = x(x+2)$. On the same axes, sketch $y = (x-a)(x-a+2)$, where $a > 2$.



Note that our intercepts are in terms of a.

Sketch $y = x^2(x-4)$. On the same axes, sketch the graph with equation

$$y = (2x)^2(2x-4)$$



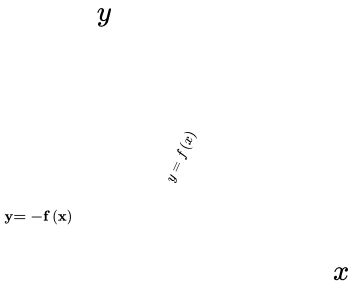
Chapter 4 · Original slide 38

Reflections of Graphs

38

If $y = x(x+2)$, sketch $y = f(x)$ and $y = -f(x)$ on the same axes.

You did this at GCSE. The minus is outside the function, so affects the output, i.e. the y value. The y values are negated, resulting in a reflection in the x-axis.



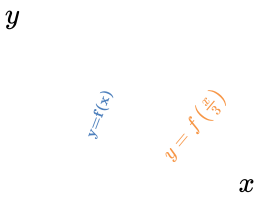
Graphs and Transformations

Chapter 4 · Original slide 39

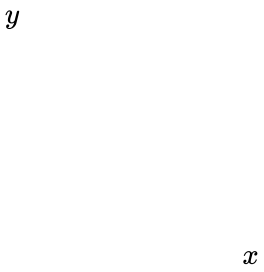
Test Your Understanding

39

If $y = (x+1)(x-2)$, sketch $y = f(x)$ and $y = f(\frac{x}{3})$ on the same axes.



Sketch the graph of $y = \frac{2}{x} + 1$, ensuring you indicate any intercepts with the axes.



Chapter 4 · Original slide 40

Exercise 4E – Pure Y1 page 74-75 (translations)

40

7 The point $P(4,-1)$ lies on the curve with equation $y = \mathrm{f}(x)$.
a State the coordinates that point P is transformed to on the curve with equation $y = \mathrm{f}(x-2)$.
b State the coordinates that point P is transformed to on the curve with equation $y = \mathrm{f}(x)+3$.

8 The graph of $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = \frac{1}{x}$ is translated so that the asymptotes are at $x = 4$ and $y = 0$. Write down the equation for the transformed function in the form $y = \frac{1}{x+a}$.

Graphs and Transformations

Chapter 4 · Original slide 41

Exercise 4E – Pure Y1 page 74-75 (translations)

41

11 a Sketch the graph of $y = x^3 + 4x^2 + 4x$. **b** The poin
coordinates $(-1, 0)$ lies on the curve with equation
$y = (x + a)^3 + 4(x + a)^2 + 4(x + a)$ where a is a constant. Find
the two possible values of a.

Chapter 4 · Original slide 42

Exercise 4E – Pure Y1 page 74-75 (translations)

42

12 a Sketch the graph of $y = x(x + 1)(x + 3)^2$. **b** Find the possible values of b such that
the point $(2, 0)$ lies on the curve with equation $y = (x + b)(x + b + 1)(x + b + 3)^2$.

Graphs and Transformations

Chapter 4 · Original slide 43

Exercise 4F – Pure 1 Page 78 (stretches/reflections)

43

7 The point $P(2, -3)$ lies on the curve with equation $y = \mathrm{f}(x)$.

a State the coordinates that point P is transformed to on the curve with equation $y = \mathrm{f}(2x)$.

b State the coordinates that point P is transformed to on the curve with equation $y = 4\mathrm{f}(x)$.

8 The point $Q(-2, 8)$ lies on the curve with equation $y = \mathrm{f}(x)$. State the coordinates that point Q is transformed to on the curve with equation $y = \mathrm{f}(\frac{1}{2}x)$.

Chapter 4 · Original slide 44

Exercise 4F – Pure 1 Page 78 (stretches/reflections)

44

9 a Sketch the graph of $y = (x - 2)(x - 3)^2$. **b** The graph of $y = (ax - 2)(ax - 3)^2$ passes through the point $(1, 0)$. Find two possible values for a.

Graphs and Transformations

Chapter 4 · Original slide 45

Effect of transformation on specific points

45

Sometimes you will not be given the original function, but will be given a sketch with specific points and features you need to transform.
Where would each of these points end up?

$\mathbf{y=}f(x)$	$(4,3)$	$(1,0)$	$(6,-4)$
$y=f(x+1)$			
$y=f(2x)$			
$y=3f(x)$			
$y=f(x)-1$			
$y=f\left(\frac{x}{4}\right)$			
$y=f(-x)$			
$y=-f(x)$			

Chapter 4 · Original slide 46

Test Your Understanding

46

Edexcel C1 May 2012
Q10

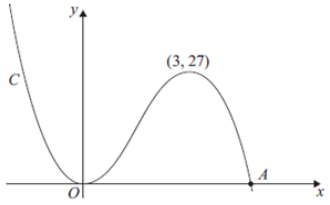


Figure 1

(a)

(b)(i)

(ii)

(c)

Graphs and Transformations

Chapter 4 · Original slide 47

Exercise 4G Pure Y1 Pages 80

47

5 The point $P(2, 1)$ lies on the graph with equation $y = \mathrm{f}(x)$.

a On the graph of $y = \mathrm{f}(ax)$, the point P is mapped to the point $Q(4, 1)$. Determine the value of a.

b Write down the coordinates of the point to which P maps under each transformation

i $\mathrm{f}(x - 4)$ ii $3\mathrm{f}(x)$ iii $\frac{1}{2}\mathrm{f}(x) - 4$

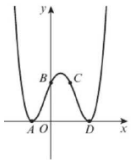
Chapter 4 · Original slide 48

Exercise 4G Pure Y1 Pages 80

48

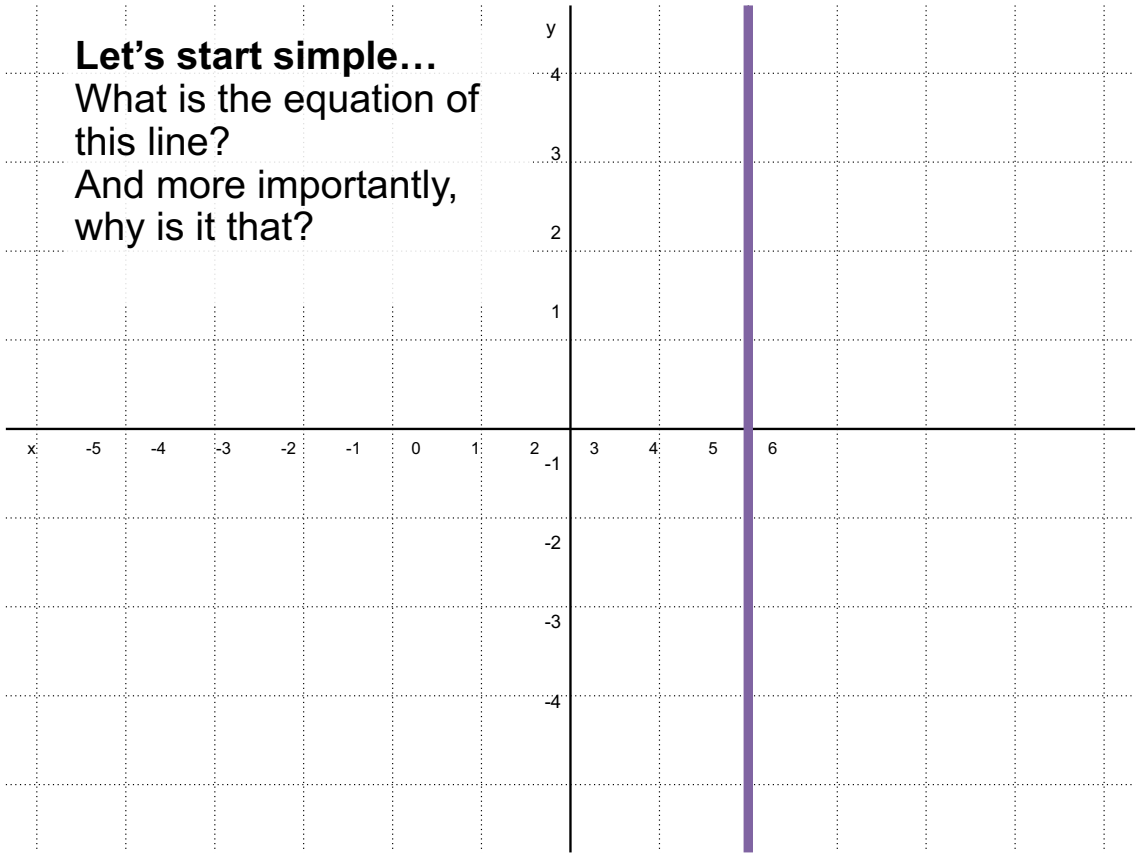
6 The diagram shows a sketch of a curve with equation $y = \mathrm{f}(x)$. The points $A(-1, 0), B(0, 2), C(1, 2), D(2, 0)$ lie on the curve. Sketch the following graphs and give the coordinates of the points A, B, C, D after each transformation:

a $y + 2 = \mathrm{f}(x)$ **b** $\frac{1}{2}y = \mathrm{f}(x)$ **c** $y - 3 = \mathrm{f}(x)$ **d** $3y = \mathrm{f}(x)$ **e** $2y - 1 = \mathrm{f}(x)$



Graphs and Transformations

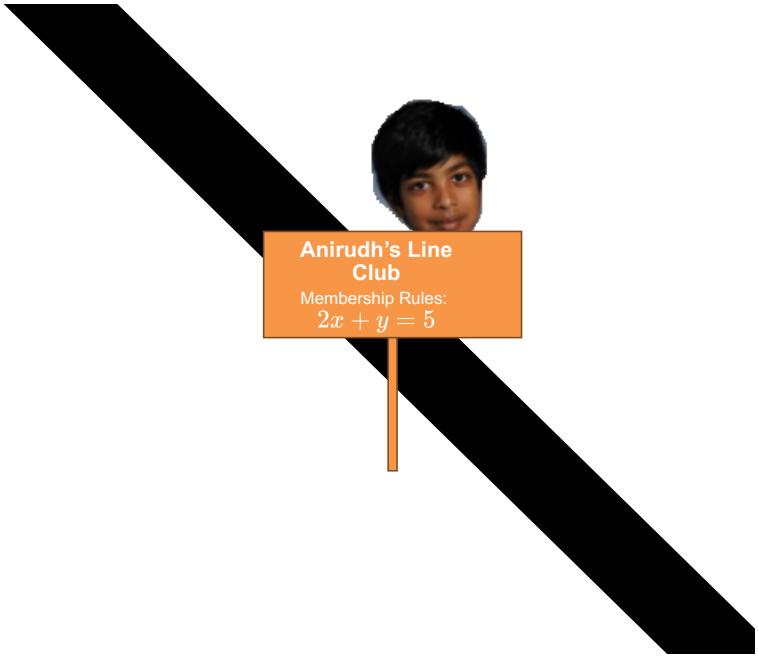
Chapter 5 · Original slide 4



Chapter 5 · Original slide 5

Lines and Equations of Lines

! A line consists of all points which satisfy some equation in terms of x and/or y.



Graphs and Transformations

Chapter 5 · Original slide 6

Examples

This means we can **substitute** the values of a coordinate into our equation whenever we know the point lies on the line.

The point $(5, a)$ lies on the line with equation $y = 3x + 2$. Determine the value of a.

Find the coordinate of the point where the line $2x + y = 5$ cuts the x-axis.

Chapter 5 · Original slide 7

Test Your Understanding

Determine where the line $x + 2y = 3$ crosses the:

a) y-axis:

b) x-axis:

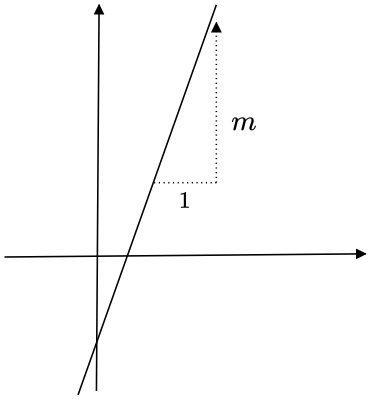
What mistakes do you think it's easy to make?

Graphs and Transformations

Chapter 5 · Original slide 8

Recap of gradient

The steepness of a line is known as the **gradient**.
It tells us what y changes by as x increases by 1.



So if the y value increased by 6 as the x value increased by 2, what is y increasing by for each unit increase of x?
How would that give us a suitable formula for the gradient m?

$$m = \frac{\Delta y}{\Delta x}$$

Chapter 5 · Original slide 9

Examples

Find the gradient of the line that goes through the points:

1. $(1, 4)$ $(3, 10)$
2. $(5, 7)$ $(8, 1)$
3. $(2, 2)$ $(-1, 10)$
4. Show that the points $A(3, 4), B(5, 5), C(11, 8)$ all lie on a straight line.

If points are 'collinear' they lie on the same line.

Graphs and Transformations

Chapter 5 · Original slide 10

Further Example

The line joining $(2, -5)$ to $(4, a)$ has gradient -1. Work out the value of a.

Chapter 5 · Original slide 11

$y = mx + c$

One form we can put a straight line equation in is:

$$y = mx + c$$

Gradient

y-intercept

Why does it work?

- **The $\mathbf{y}$-intercept by definition is the $\mathbf{y}$ value when $\mathbf{x=0}$.**
  Substituting:
  $\mathbf{y=(m\times 0)+c=c}$ **as expected.**
- **By the definition of gradient, if we increase $\mathbf{x}$ by 1, $\mathbf{y}$ should increase by $\mathbf{m}$:**
  $\mathbf{y=m(x+1)+c=mx+m+c}$
  which indeed has increased by $\mathbf{m}$.

Graphs and Transformations

Chapter 5 · Original slide 12

$y = mx + c$

Determine the gradient and y-intercept of the line with equation $4x - 3y + 5 = 0$

Make y the subject so we have the form $y = mx + c$
Put y on the side it's positive.

Chapter 5 · Original slide 13

$ax + by + c = 0$

At GCSE, $\mathbf{y=mx+c}$ was the main form you would express a straight line equation, sometimes known as the '**slope-intercept form**'.
But another common form is $\mathbf{ax+by+c=0}$, where a, b, c are integers.
This is known as the '**standard**' form.

Express $y = \frac{1}{3}x - \frac{2}{3}$ in the form $ax + by + c = 0$, where a, b, c are integers.

We don't want fractions, so multiply by an appropriate number.

Put everything on either side of equation.
$-x + 3y + 2 = 0$ would also be OK.

We'll see on the next slide WHY we might want to put an equation in this form over $y = mx + c$...

Graphs and Transformations

Chapter 5 · Original slide 14

Just for your interest…

Why might we want to put a straight line equation in the form $ax + by + c = 0$?

$y = mx + c$
"Slope-Intercept Form"

$ax + by + c = 0$
"Standard Form"

Coverage

$y = mx + c$ doesn't allow you to represent vertical lines. Standard form allows us to do this by just making b zero.

$$x + 4 = 0$$

Symmetry

In general, the '**linear combination**' of two variables x and y is $ax + by$, i.e. "some amount of x and some amount of y". There is a greater elegance and symmetry to this form over $y = mx + c$ because x and y appear similarly within the expression.

Usefulness

This more 'elegant' form also means it ties in with vectors and matrices. In FM, you will learn about the '**dot product**' of two vectors:

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = ax + by$$

thus since $ax + by + c = 0$, we can represent a straight line using:

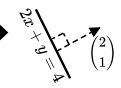
$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} + c = 0 \qquad (1)$$

We can extend to 3D points to get the equation of a **plane**:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} + d = 0 \qquad (2)$$

Conveniently, in equation (1), the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is **perpendicular to the line**. And in equation (2), the vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is perpendicular to the plane. Nice!

$$2x + y = 4 \quad \Rightarrow \quad \begin{pmatrix} 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = 4$$



Chapter 5 · Original slide 15

Test Your Understanding

Express $y = \frac{2}{5}x + \frac{3}{5}$ in the form $ax + by + c = 0$, where a, b, c are integers.

Graphs and Transformations

Chapter 5 · Original slide 16

Exercise 5A – Pure Y1 Page 91

9 Show that the points $A(2, 3)$, $B(4, 4)$ and $C(10, 7)$ can be joined by a straight line.

10 Show that the points $A(-2a, 5a)$, $B(0, 4a)$ and points $C(6a, a)$ are collinear. $(3\ \mathrm{ma}\ldots$

Chapter 5 · Original slide 17

Exercise 5B – Pure Y1 Page 92-93

8 A line l cuts the x-axis at $(5, 0)$ and the y-axis at $(0, 2)$.

a Find the gradient of the line.

b Find an equation of the line in the form $ax + by + c = 0$.

9 Show that the line with equation $ax + by + c = 0$ has gradient $-\frac{a}{b}$ and cuts the y-axis at $-\frac{c}{b}$.

10 The line l with gradient 3 and y-intercept $(0, 5)$ has the equation $ax - 2y + c = 0$.

Find the values of a and c. (2 marks)

Graphs and Transformations

Chapter 5 · Original slide 18

Exercise 5B – Pure Y1 Page 92-93

11 The straight line l passes through $(0, 6)$ and has gradient -2. It intersects the line with equation $5x - 8y - 15 = 0$ at point P. Find the coordinates of P.

12 The straight line l_1 with equation $y = 3x - 7$ intersects the straight line l_2 with equation $ax + 4y - 17 = 0$ at the point $P(-3, b)$.

a Find the value of b.

b Find the value of a.

Chapter 9 · Original slide 44

Sin Graph

What does it look like?

Key Features:
- Repeats every $360°$.
- Range: $-1 < \sin(x) < 1$

-360 -270 -180 -90 90 180 270 360

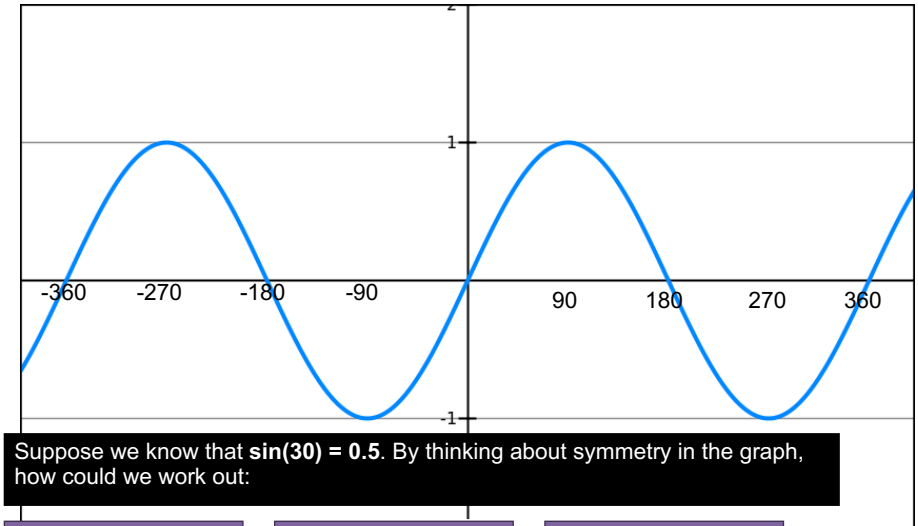
Graphs and Transformations

Chapter 9 · Original slide 45

Sin Graph

45

What do the following graphs look like?



Suppose we know that **sin(30) = 0.5**. By thinking about symmetry in the graph, how could we work out:

sin(150) =

sin(-30) =

sin(210) =

Chapter 9 · Original slide 46

Cos Graph

46

What do the following graphs look like?

Key Features:
- Repeats every $360°$.
- Range: $-1 < \sin(x) < 1$

-360 -270 -180 -90 90 180 270 360

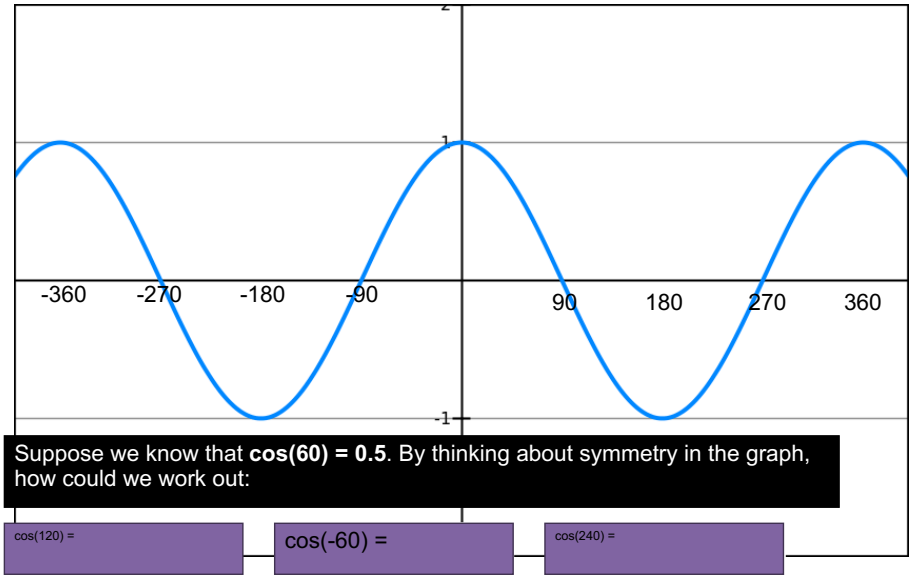
Graphs and Transformations

Chapter 9 · Original slide 47

Cos Graph

47

What does it look like?



Suppose we know that **cos(60) = 0.5**. By thinking about symmetry in the graph, how could we work out:

cos(120) =

cos(-60) =

cos(240) =

Chapter 9 · Original slide 48

Tan Graph

48

What does it look like?

-360 -270 -180 -90 90 180 270 360

Key Features:
- Repeats every $\mathbf{180}^\circ$.
- Roots: $0, 180^\circ, -180^\circ, \ldots$
- Range: $\tan(x) \in \mathbb{R}$ (i.e. no min/ max value!)
- Asymptotes: $x = \pm 90^\circ, \pm 270^\circ, \ldots$

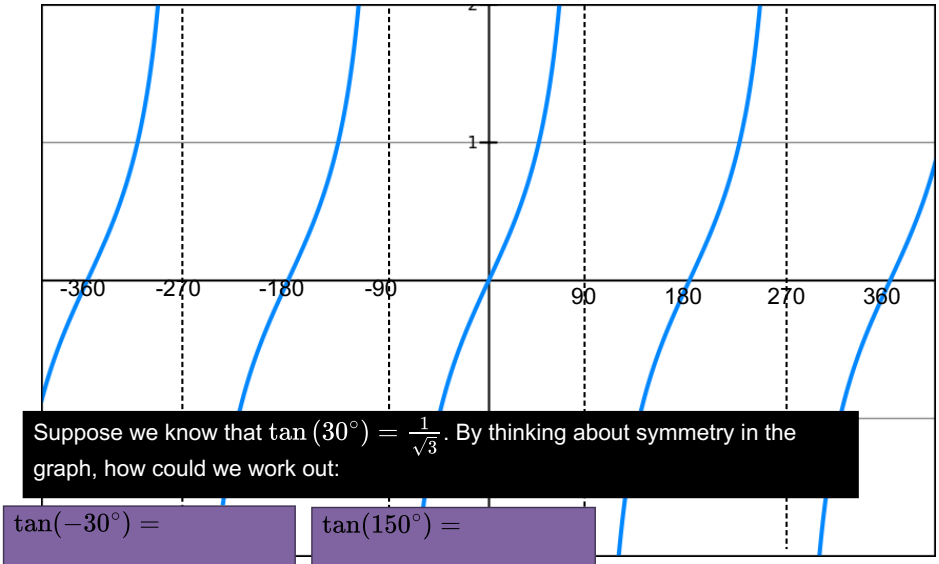
Graphs and Transformations

Chapter 9 · Original slide 49

Tan Graph

49

What does it look like?



Suppose we know that $\tan(30^\circ) = \frac{1}{\sqrt{3}}$. By thinking about symmetry in the graph, how could we work out:

$\tan(-30^\circ) =$

$\tan(150^\circ) =$

Chapter 9 · Original slide 50

Transforming Trigonometric Graphs

50

There is no new theory here: just use your knowledge of transforming graphs, i.e. whether the transformation occurs 'inside' the function (i.e. input modified) or 'outside' the function (i.e. output modified).

Sketch $y = 4\sin x,\ 0 \leq x \leq 360^\circ$

y

90° 180° 270° 360° x -90°

Sketch $y = \cos(x + 45^\circ),\ 0 \leq x \leq 360^\circ$

y

90° 180° 270° 360° x

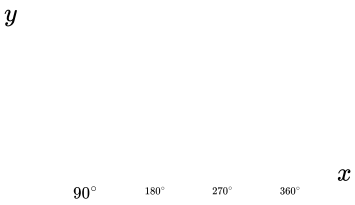
Graphs and Transformations

Chapter 9 · Original slide 51

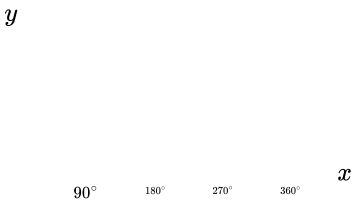
Transforming Trigonometric Graphs

51

Sketch $y = -\tan x,\ 0 \leq x \leq 360^\circ$



Sketch $y = \sin\left(\frac{x}{2}\right),\ 0 \leq x \leq 360^\circ$



Chapter 9 · Original slide 52

Exercise 9F – Y1 Page 194

52

1 Sketch the graph of $y = \cos\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.\par2 Sketch the graph of $y = \tan\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.\par3 Sketch the graph of $y = \sin\theta$ in the interval $-90^\circ \leq \theta \leq 270^\circ$.

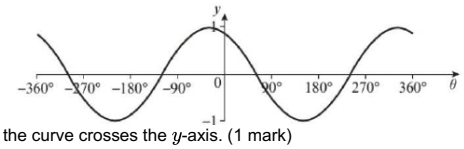
Graphs and Transformations

Chapter 9 · Original slide 53

Exercise 9G – Y1 Pages 197-198

53

7 The graph shows the curve
$y = \cos(x + 30^\circ), -360^\circ \leq x \leq 360^\circ$.
a Write down the coordinates of the points where the curve crosses the x-axis. (2 marks)



b Find the coordinates of the point where the curve crosses the y-axis. (1 mark)

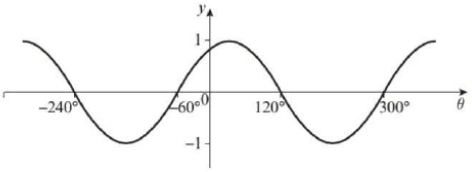
b $\cos 30^\circ = \frac{\sqrt{3}}{2}$; $\left(0, \frac{\sqrt{3}}{2}\right)$

Chapter 9 · Original slide 54

Exercise 9G – Y1 Pages 197-198

54

8 The graph shows the curve with equation $y = \sin(x + k), -360^\circ \leq x \leq 360^\circ$, where k is a constant.
a Find one possible value for k. (2 marks)
b Is there more than one possible answer to part a? Give a reason for your answer. (2 marks)



Graphs and Transformations

Chapter 9 · Original slide 55

Exercise 9G – Y1 Pages 197-198

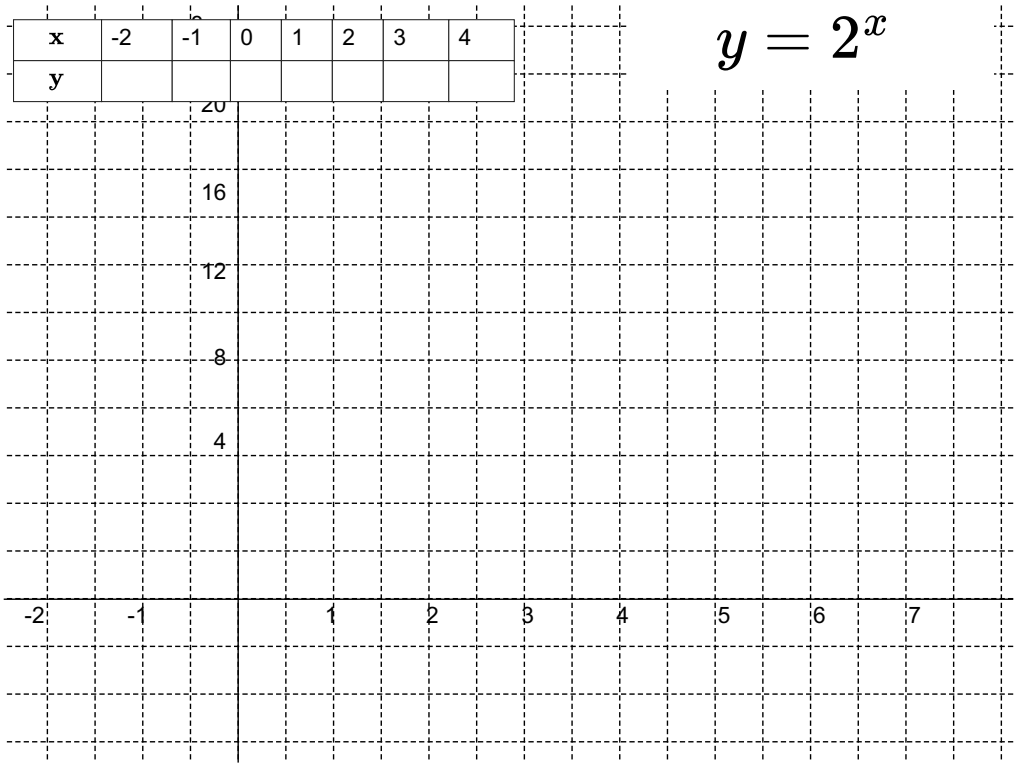
55

9 The variation in the depth of water in a rock pool can be modelled using the function $y = \sin(30t)^\circ$, where t is the time in hours and $0 \leq t \leq 6$.
a Sketch the function for the given interval.
b If $t = 0$ represents midday, during what times will the rock pool be at least half full?

Chapter 14 · Original slide 4

x	-2	-1	0	1	2	3	4
y							

$y = 2^x$



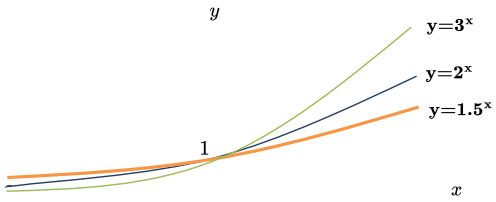
Graphs and Transformations

Chapter 14 · Original slide 5

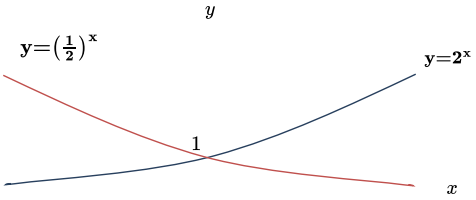
Contrasting exponential graphs

5

On the same axes sketch $y = 3^x, y = 2^x, y = 1.5^x$



On the same axes sketch $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$



Three important notes:

- $y = 2^x$ is said to be "**exponential growing**" whereas $y = \left(\frac{1}{2}\right)^x$ is said to be "**exponentially decaying**", because it's getting smaller (halving) each time x increases by 1.
- $y = \left(\frac{1}{2}\right)^x$ is a reflection of $y = 2^x$ in the line $x = 0$.
  Proof:
  If $f(x) = 2^x$, $f(-x) = 2^{-x} = \frac{1}{2^x} = \left(\frac{1}{2}\right)^x$
- $\left(\frac{1}{2}\right)^x$ would usually be written 2^{-x}.

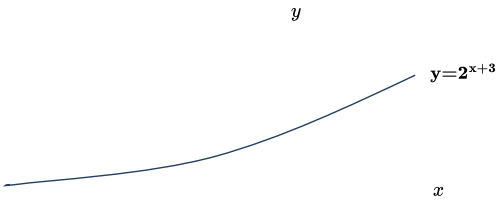
You should therefore in general be able to recognise and sketch the graph $y = a^{-x}$.

Chapter 14 · Original slide 6

Graph Transformations

6

Sketch $y = 2^{x+3}$



The 'change' to 2^x is 'inside the function' (i.e. input x is replaced with $x + 3$).

Ensure you work out the new y-intercept.

Graphs and Transformations

Chapter 14 · Original slide 7

Exercise 14A (Pages 313-314)

7

6 The graph of $y = ka^x$ passes through the points $(1, 6)$ and $(4, 48)$. Find the values of the constants k and a.

7 The graph of $y = pq^x$ passes through the points $(-3, 150)$ and $(2, 0.048)$.

a By drawing a sketch or otherwise, explain why $0 < q < 1$.

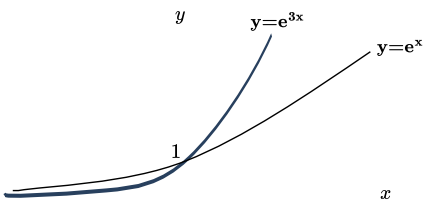
b Find the values of the constants p and q.

Chapter 14 · Original slide 11

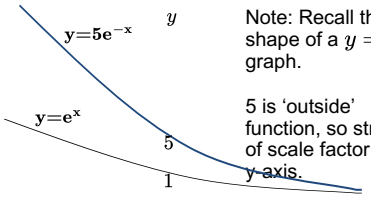
More Graph Transformations

11

Sketch $y = e^{3x}$



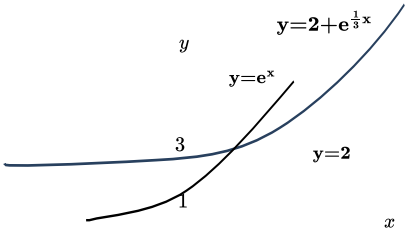
Sketch $y = 5e^{-x}$



Note: Recall the shape of a $y = a^{-x}$ graph.

5 is 'outside' function, so stretch of scale factor 5 on y-axis.

Sketch $y = 2 + e^{\frac{1}{3}x}$



Important Note: Because the original asymptote was $y = 0$, it is now $y = 2$ and you must indicate this along with its equation.

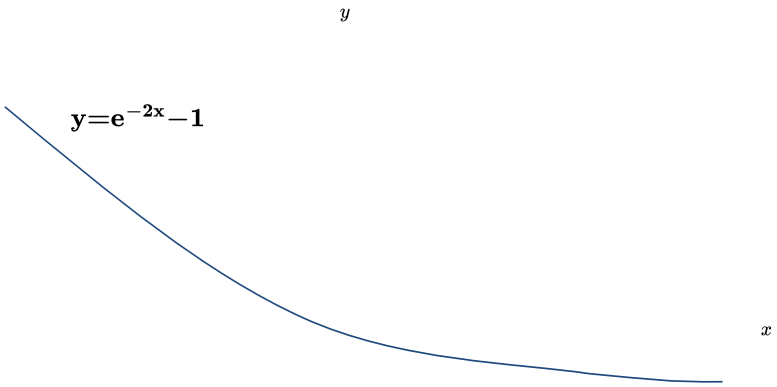
Graphs and Transformations

Chapter 14 · Original slide 12

Test Your Understanding

12

Sketch $y = e^{-2x} - 1$

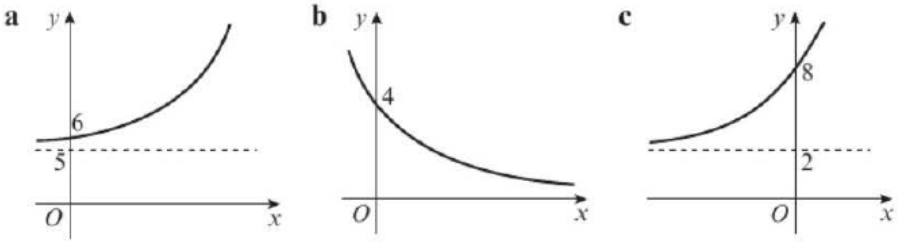


Chapter 14 · Original slide 13

Exercise 14B (Pages 316-317)

13

4 Each of the sketch graphs below is of the form $y = Ae^{bx} + C$, where A, b and C are constants. Find the values of A and C for each graph, and state whether b is positive or negative.



Graphs and Transformations

Chapter 14 · Original slide 14

Exercise 14B (Pages 316-317)

14

6 Differentiate the following with respect to x.

a e^{6x} \qquad **b** $e^{-x/3}$ \qquad **c** $7e^{2x}$

d $5e^{0.4x}$ \qquad **e** $e^{3x} + 2e^{x}$ \qquad **f** $e^{x}(e^{x} + 1)$

Chapter 14 · Original slide 15

Exercise 14B (Pages 316-317)

15

7 Find the gradient of the curve with equation $y = e^{3x}$ at the point where

a $x = 2$ **b** $x = 0$ **c** $x = -0.5$

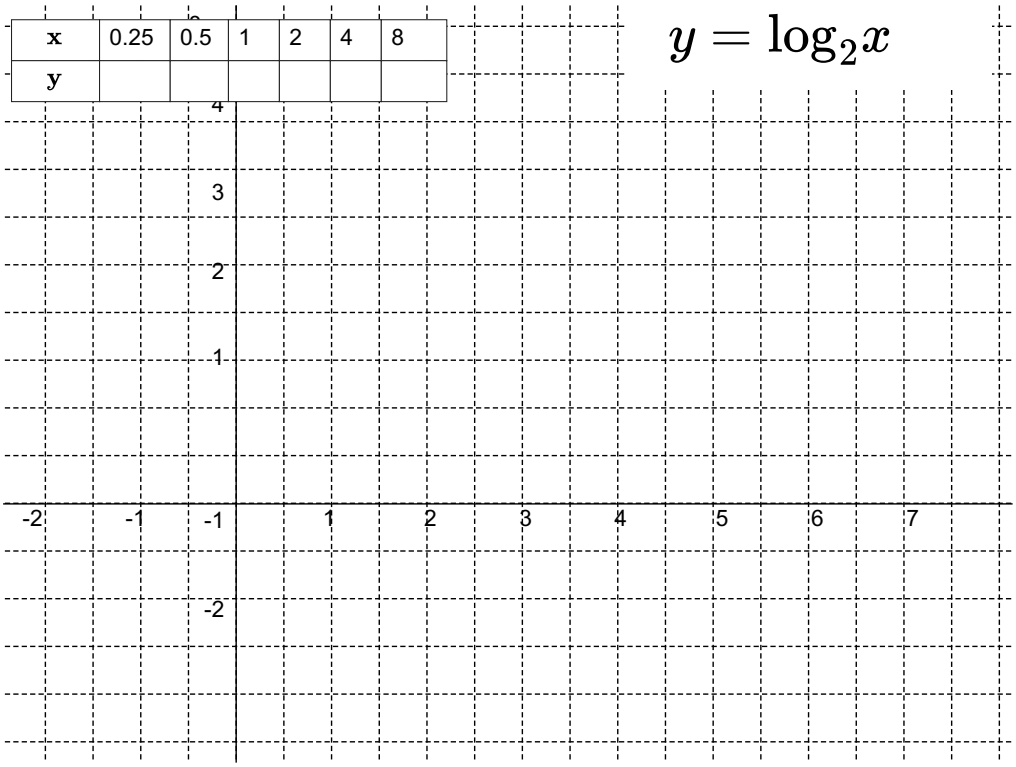
8 The function f is defined as $f(x) = e^{0.2x}$, $x \in \mathbb{R}$. Show that the tangent to the curve at the point $(5, e)$ goes through the origin.

Graphs and Transformations

Chapter 14 · Original slide 27

x	0.25	0.5	1	2	4	8
y						

$y = \log_2 x$



Chapter 14 · Original slide 41

Natural Logarithms

41

We have previously seen that $y = \log_a x$ is the inverse function of $y = a^x$.
We also saw that e^x is "**the**" exponential function.
The inverse of e^x is $\log_e x$, but because of its special importance, it has its own function name!

The inverse of $y = e^x$ is $y = \ln x$

$\ln e^x =$

$e^{\ln x} =$

Solve $e^x = 5$

"ln both sides".
On the LHS it cancels out the "e to the power of"

Solve $2 \ln x + 1 = 5$

Do "e to the power of" each side. On the LHS it cancels out the ln.